



**Thompson Bishop Sparks
State Diagnostic Laboratory
Policy or Operating Procedure**

SOP Title: Procedure for Spot Oxidase Test

Author: Christy Williams

SOP#: BAC.224.4

Area: Bacteriology

Date: 2/23/18

Rev Date: 3/13/20

Rev Level: 4

Purpose and application of procedure: The oxidase test is based on the production of an enzyme called indophenol oxidase which in the presence of atmospheric oxygen oxidizes phenylenediamine oxidase reagent to form a dark-purple compound, indophenol.

Procedure:

MASTER

I. Personal Protective Equipment

- A. Gloves
- B. Laboratory coat
- C. Biological safety cabinet (as needed)

II. Equipment

- A. Sterile cotton swabs

III. Reagents

- A. Oxidase reagent

IV. Test Procedure

- A. For identification testing, the use of a biological safety cabinet is not necessary unless the biological safety levels of the suspected organism require the use of a biological safety cabinet.
- B. Hold the reagent dropper upright and point away from yourself. Grasp the middle with thumb and forefinger and squeeze gently to crush the ampule inside the dropper.
- C. Tap the bottom on the benchtop a few times. Then invert for dispensing of reagent.
- D. Using a cotton swab, pick up growth from the culture.
- E. Add a few drops of Oxidase reagent to the swab.
- F. Observe for color change. Test results should be read within 30 seconds of addition of reagent.
- G. A color change to dark purple constitutes a positive reaction. Negative reactions remain colorless or turn light pink. Delayed reactions should be ignored.
- H. If results are not as expected, test the reagent with *Pseudomonas aeruginosa* ATCC 27853 and *Escherichia coli* ATCC 25922.
- I. Record test results on the bacteriology/mycology worksheet.

V. Disposal

Testing supplies should be discarded in biological waste bins placed throughout the bacteriology laboratory. Waste is processed as outlined in BAC.232
Procedure for Disposal of Laboratory Waste.

Approved:

Date: 3/29/20

QM Approved:

Date: 3/30/20

			
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Associated Forms or Files:

N/A

Associated SOPs:

BAC.232 Procedure for Disposal of Laboratory Waste

Revisions:

Added printing the COA for QC.

Corrected typo in reference from 02/2015 to 05/2015.

References:

BD Oxidase Reagent Droppers Package Insert L001133(02), 05/2015.

Quality Control:

Print certificate of analysis for each lot number and place in the COA notebook.

Policy Implementation steps:

N/A